

EXPERIMENT NO. 02

TITLE: SWITCHING MODULATOR

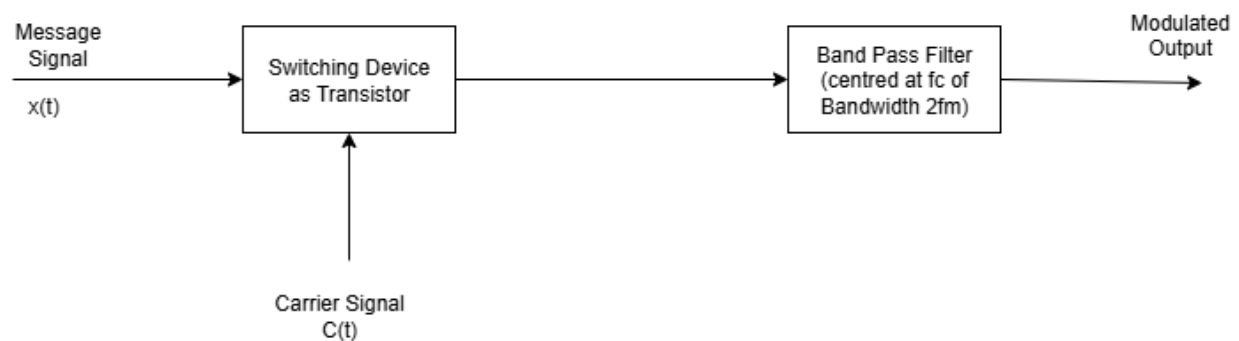


**EC39001: Communication I Lab
Year: 2026
Department of E&ECE
IIT Kharagpur**

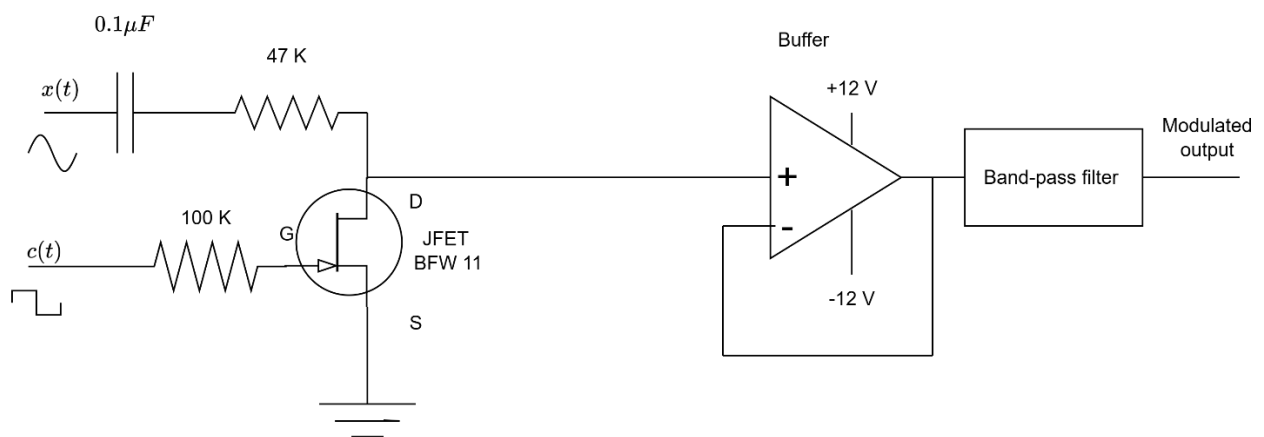
Theory:

A high-speed electronic switch, usually a transistor, is used in a switching modulator to respond to the carrier signal by turning on and off. The message signal can flow (or be multiplied) when the switch is ON; it can be blocked when it is OFF. As a result, the carrier and the message signal behave like they are multiplying.

Switching Modulator using a Transistor:



Circuit Diagram:



$x(t)$ = Message signal (Sine wave, $4V_{pp}$, $f_m=1KHz$),
 $c(t)$ = Carrier signal (Square wave, Duty cycle= 50%, $f_c=10KHz$)

- By using the transfer function of the BPF, find the following:
i) Mid frequency f_m , ii) Gain at f_m , iii) Filter Quality(Q), iv) Bandwidth(B).

Note: You may use a second-order multiple feedback type BPF.

Expected output:

